

Erdős Problem #1060

EP-1060

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $f(n)$ count the number of solutions to $k\sigma(k) = n$, where $\sigma(k)$ is the sum of divisors of k . Is it true that $f(n) \leq n^{o(\frac{1}{\log \log n})}$? Perhaps even $\leq (\log n)^{O(1)}$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>